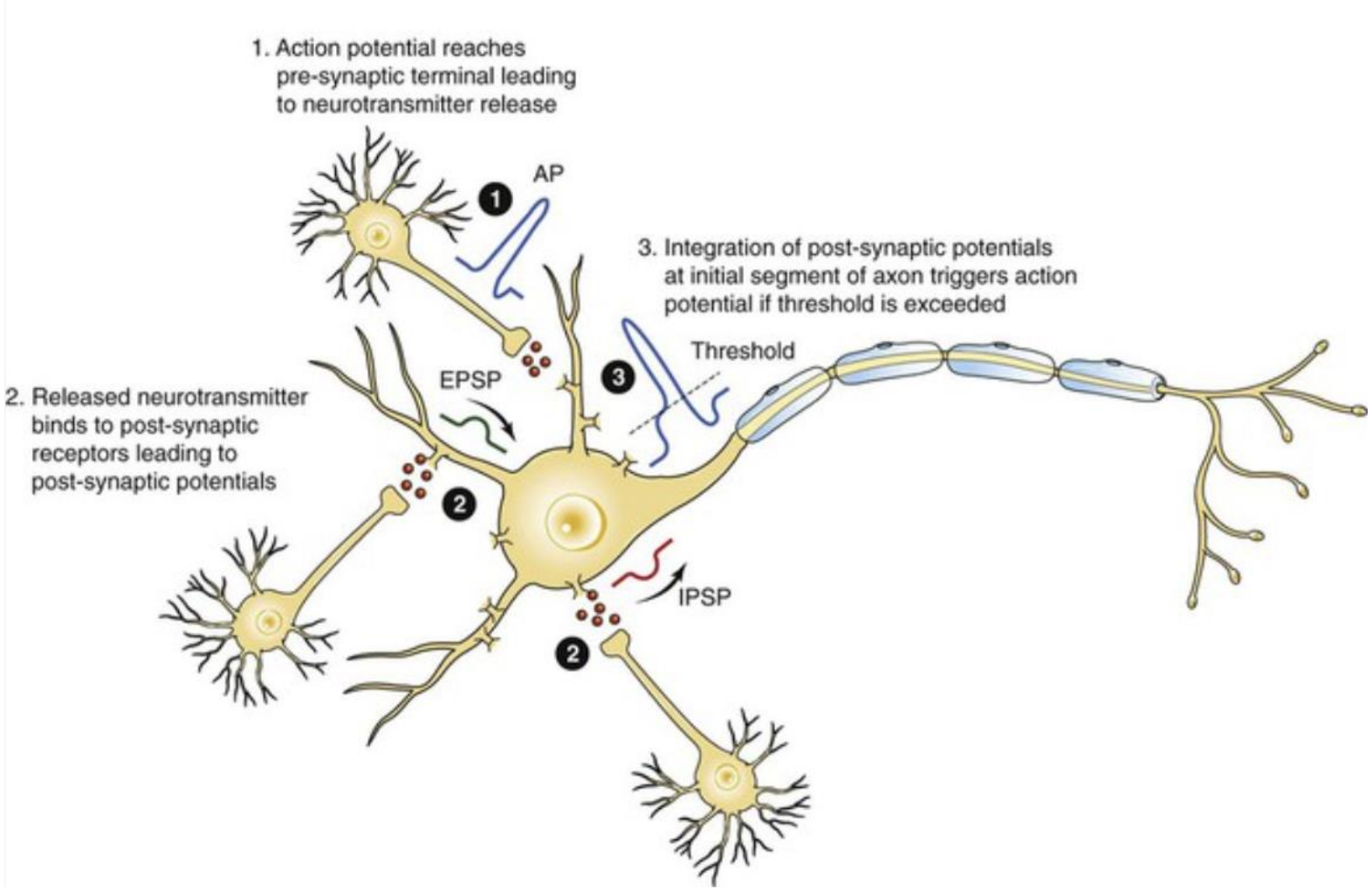
Motivation, and canonical method, to analyze “neural populations”:

Principal Component Analysis

Laureline Logiaco

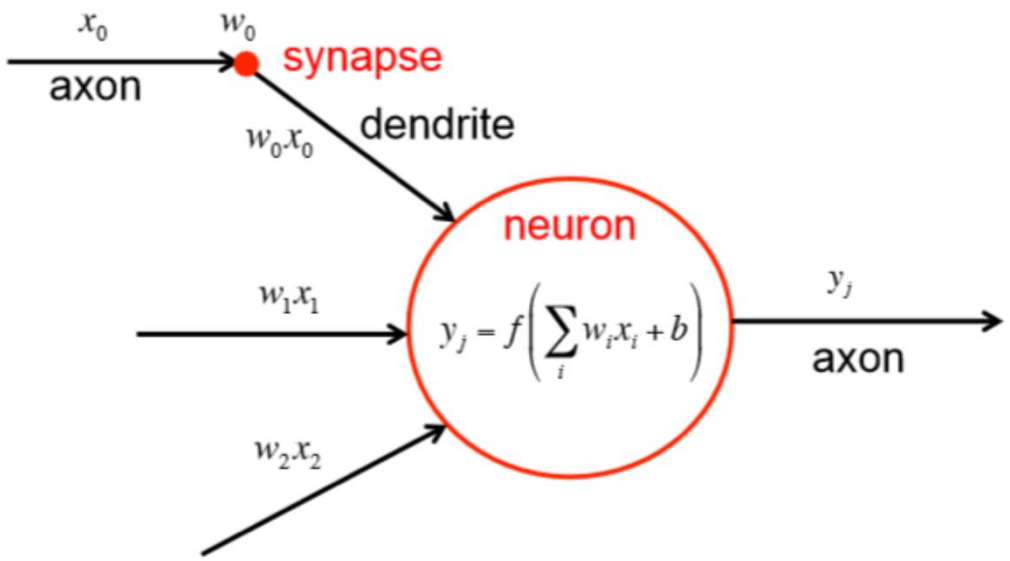
Motivation: neurons, and downstream muscles, integrate many synaptic inputs

Receptors= ‘weights’



https://veteriankey.com/the-neuron/

How to discover neurons that may ‘act together’- and how to find a small number of activity patterns?



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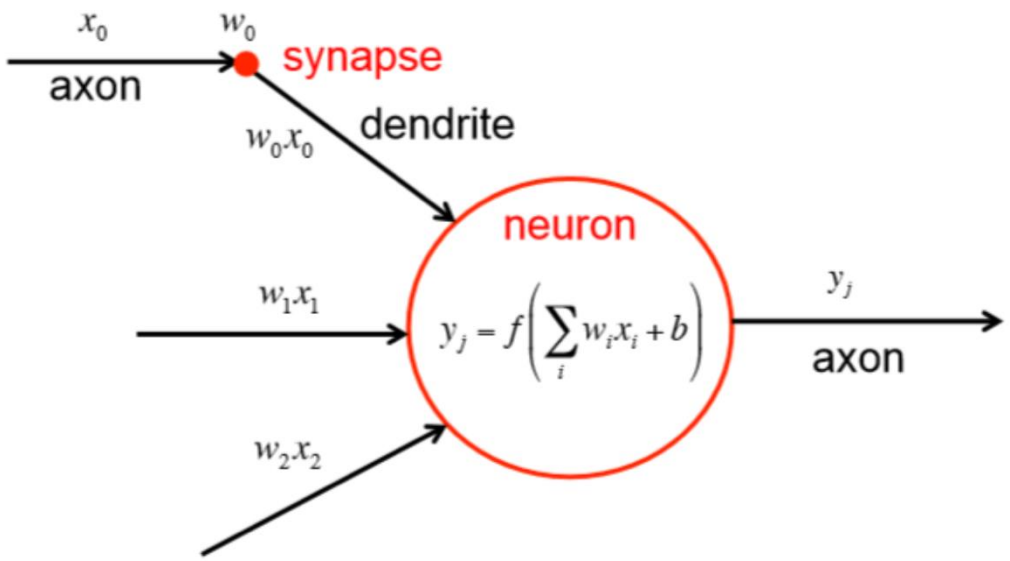

From N neurons; unknown weights, unknown identities:

find “main” k<N weight patterns $\{w_i\}_{\text{i in [1, …, N]}}$

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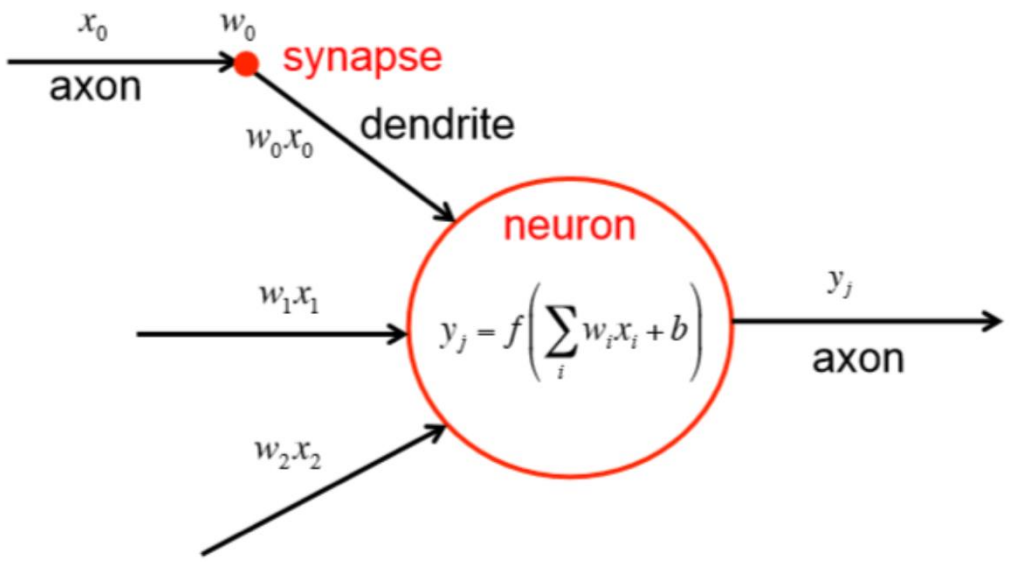

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find “main” k<N weight patterns $\{w_i\}_{\text{i in [1, …, N]}}$

<u>Principal Component Analysis (PCA):</u>

- weights that lead to most variation across time/trials, capturing covariations between neurons
- weights that ”explain/reconstruct” most of the population activity

Principal component analysis: PCA

Data: X

Features, e.g. neurons

Observations,
e.g. binned firing rates in different
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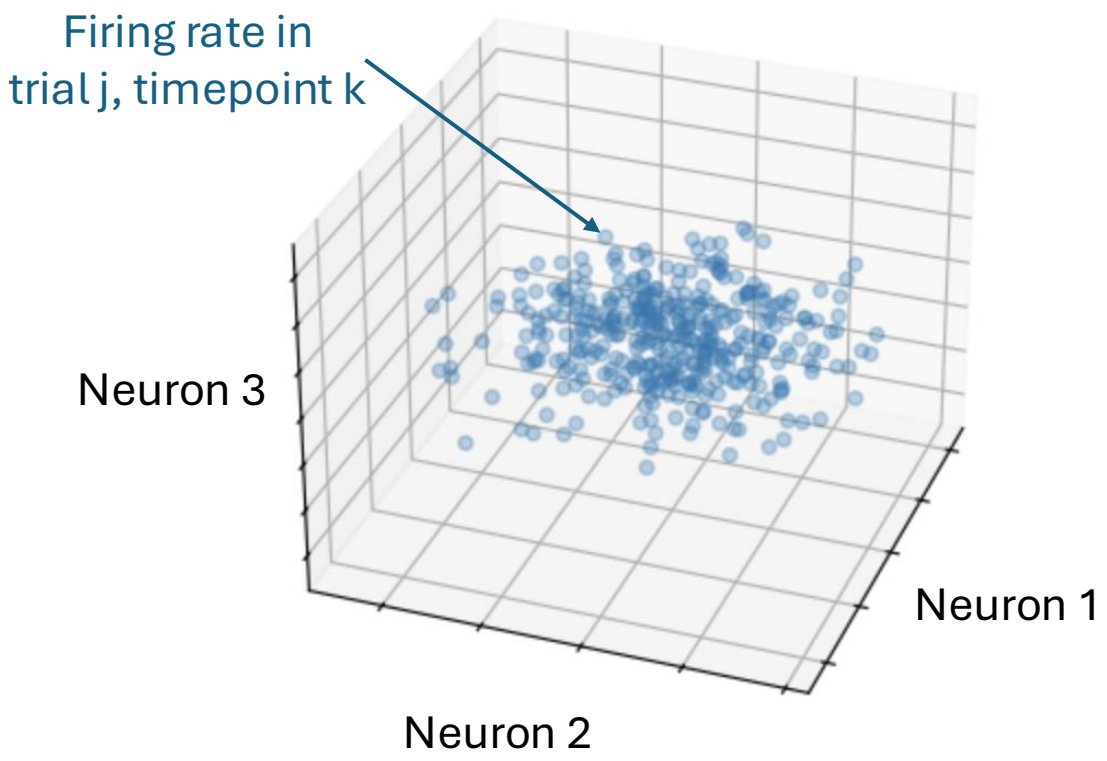

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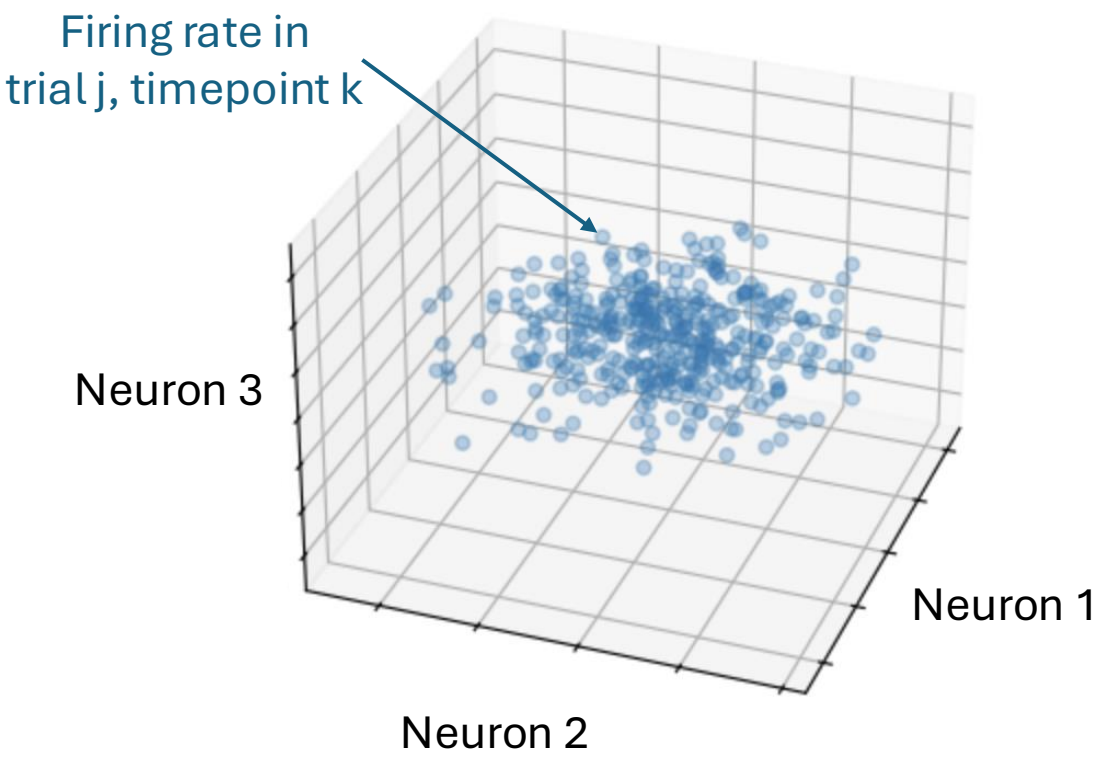

PCA tries to find a few directions / axes / set of weights for the activity that capture most of the data

Principal component analysis: PCA

<u>Step 1</u>: center the data: $X \in \mathbb{R}^{n \times d}$ ➡ compute feature-wise means: $\mu_j = \frac{1}{n} \sum_{i=1}^{n} X_{ij}$

Centered data X_c where $\{X_c\}_{i,j} = X_{i,j} - \mu_j$

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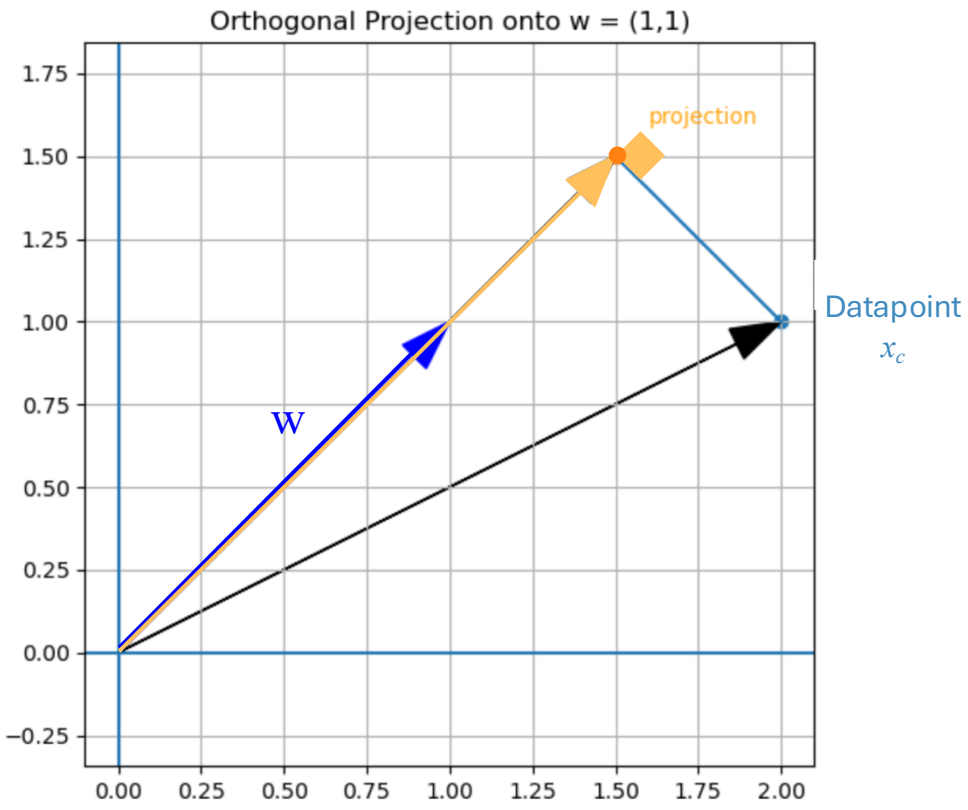

Cell 2 of notebook!

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For general w, the projection of x_c on w (yellow arrow) is:

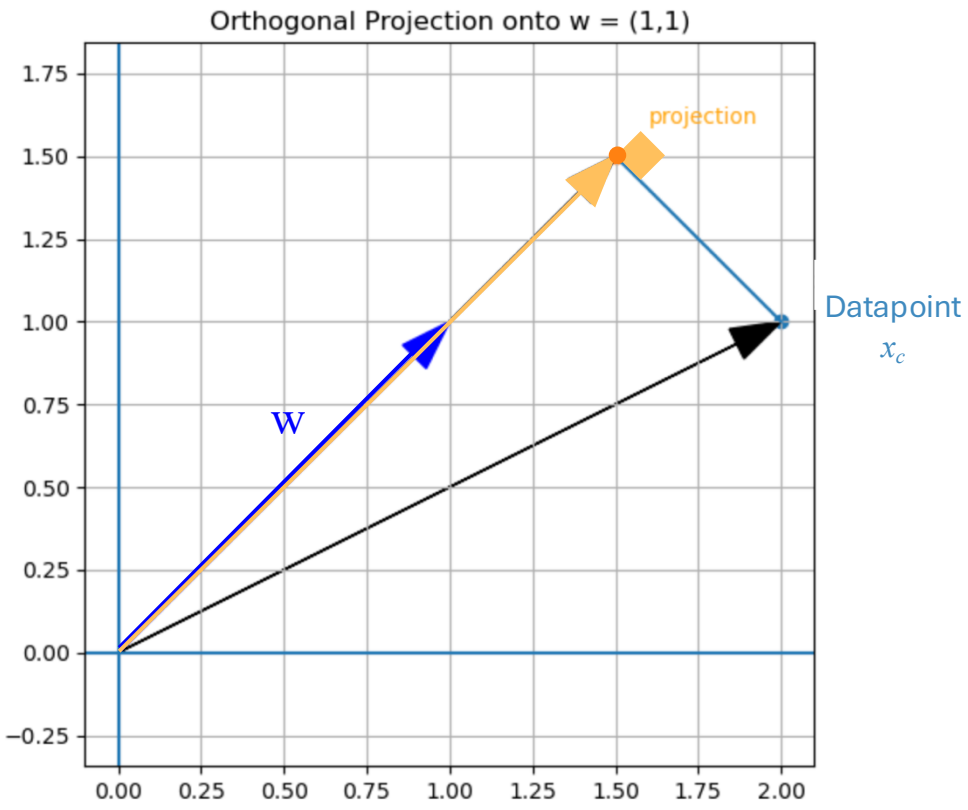
$$\text{proj}_w(x_c) = \frac{x_c^T w}{w^T w} w$$

In the figure: $x_c = (2, 1) \quad w = (1, 1)$

$$x_c^T w = x_1 w_1 + x_2 w_2 = 2 * 1 + 1 * 1 = 3$$

$$w^T w = 1^2 + 1^2 = 2$$

$$\boxed{\text{projection} = \frac{3}{2}(1, 1) = (1.5, 1.5)}$$



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<u>Step 2</u>: let w be a weight vector with norm 1 ($\sum_{i=1}^{d} w_i^2 = 1$)

For a norm-1 w, the projection of a single point x_c on w is: $\mathrm{proj}_w(x_c) = \frac{x_c^T w}{w^T w} w = (x_c^T w) w$

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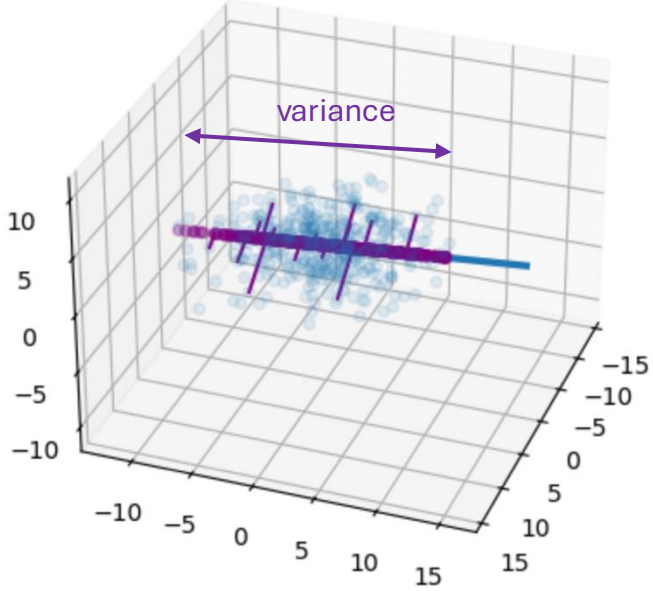
<u>Step 3:</u> <u>Find weight vectors / directions / principal axes that maximize the variance of projection magnitude:</u>

Variance (=squared magnitude for *centered* data) for several points X_c : $\frac{1}{n} \; w^T X_c^T X_c w$



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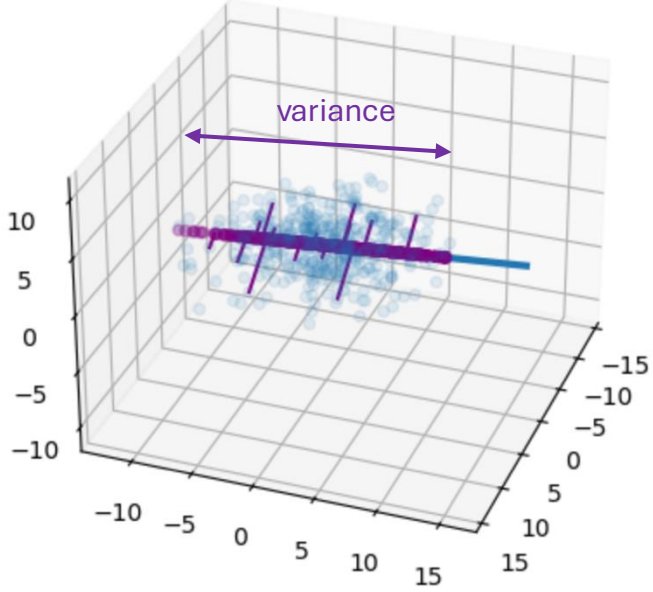

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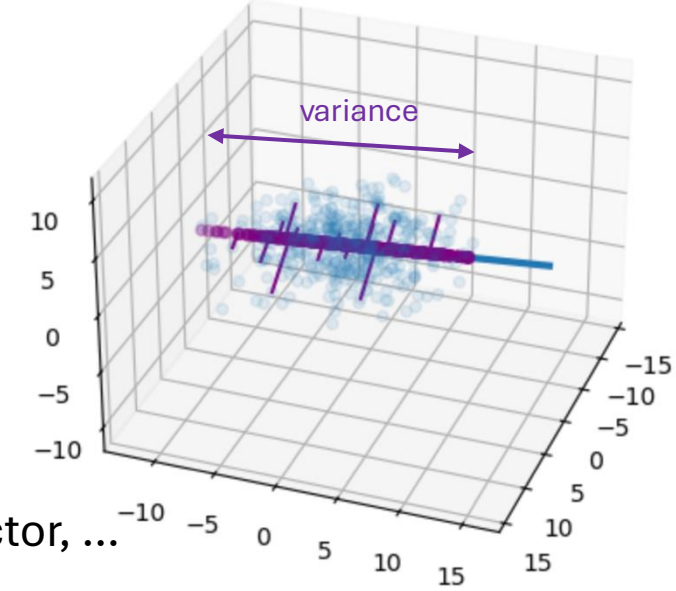
Note: optimization corresponds to eigendecomposition of Σ :

$$\underbrace{\Sigma}_{\text{covariance}} \underbrace{w}_{\text{direction}} = \underbrace{\lambda}_{\text{variance}} \underbrace{w}_{\text{same direction}}$$

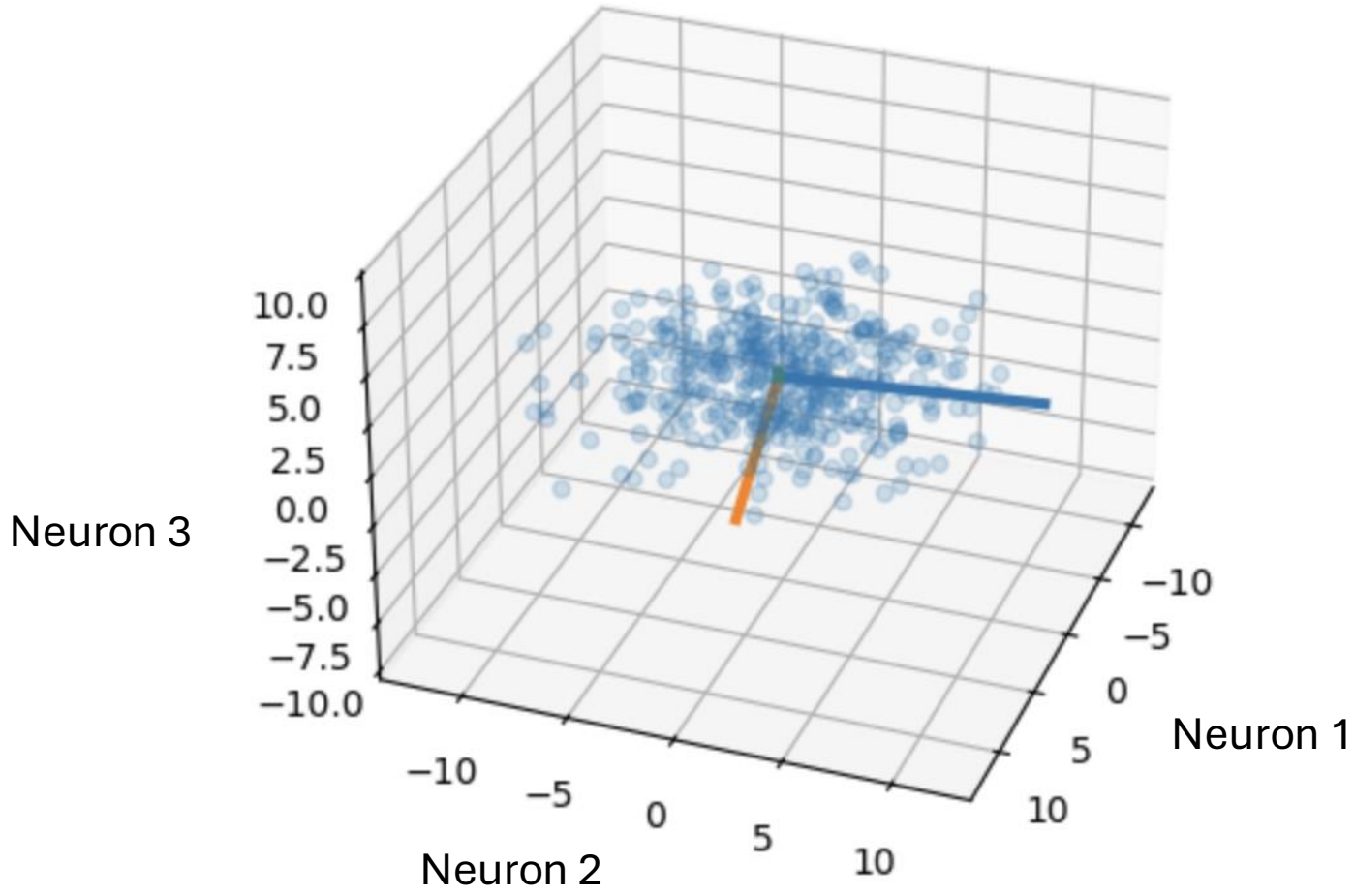


First principal axis is eigenvector with maximal eigenvalue, next axis is following eigenvector, ...

Principal component analysis: PCA

Illustrating the principal axes = orthogonal set of directions where projections have maximum variance

Principal axes (blue, orange, green) weighted by data projection variance



Cell 5 of notebook!

The first few principal axes can be used to reconstruct/denoise data!

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Use all principal axes w_i:

$$\boxed{x_c = \sum_{i=1}^{3} (\underbrace{x_c^T w_i}_{\text{Projection on } w_i})\, w_i}$$

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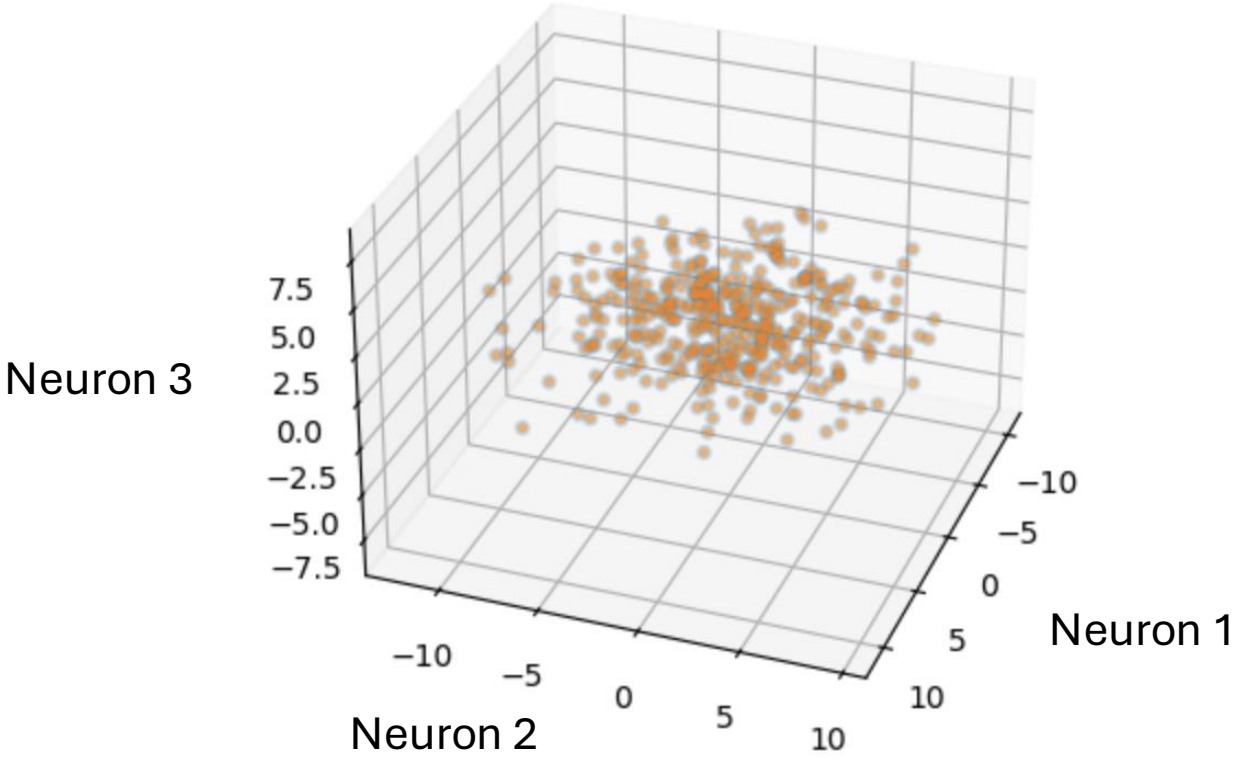

Cell 9 of notebook!

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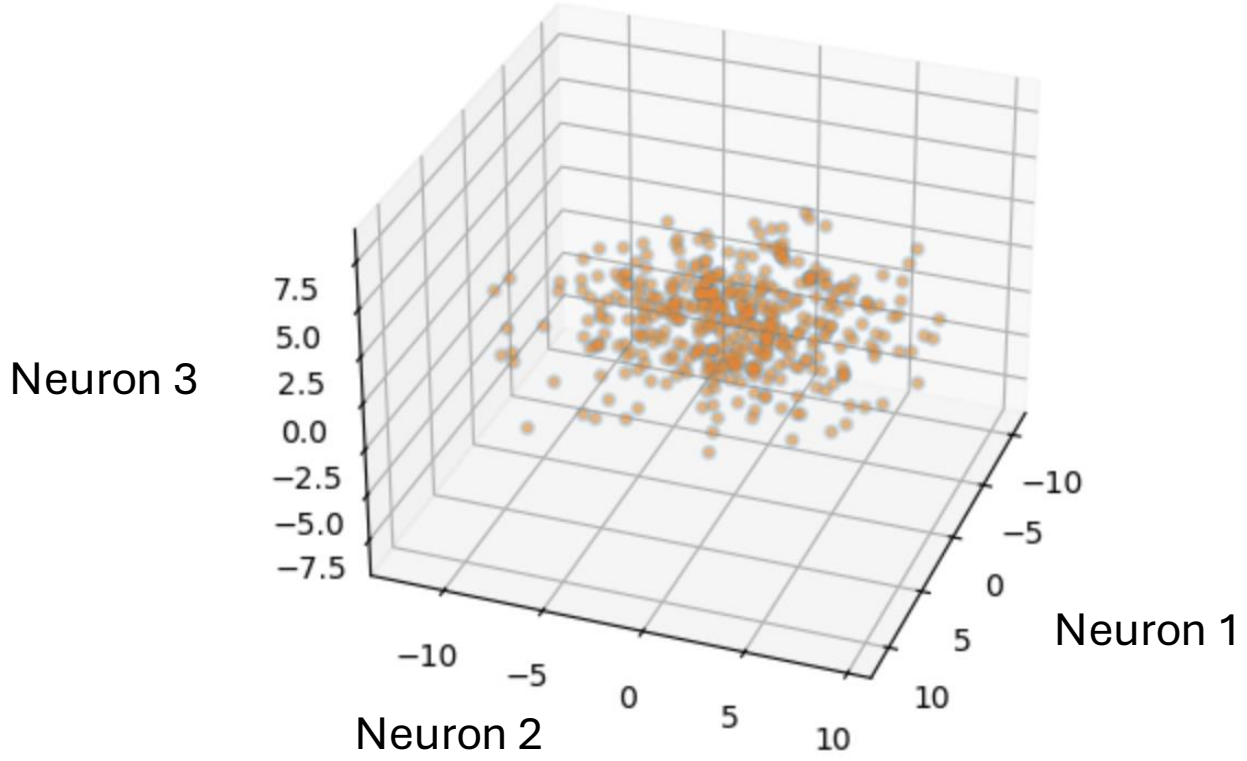

The data is perfectly reconstructed!

Cell 9 of notebook!

The first few principal axes can be used to reconstruct/denoise data!

Use only the first two axes w_i:

$$\boxed{\hat{x}_c = \sum_{i=1}^{2} (\underbrace{x_c^T w_i}_{\text{Projection on } w_i})\, w_i}$$

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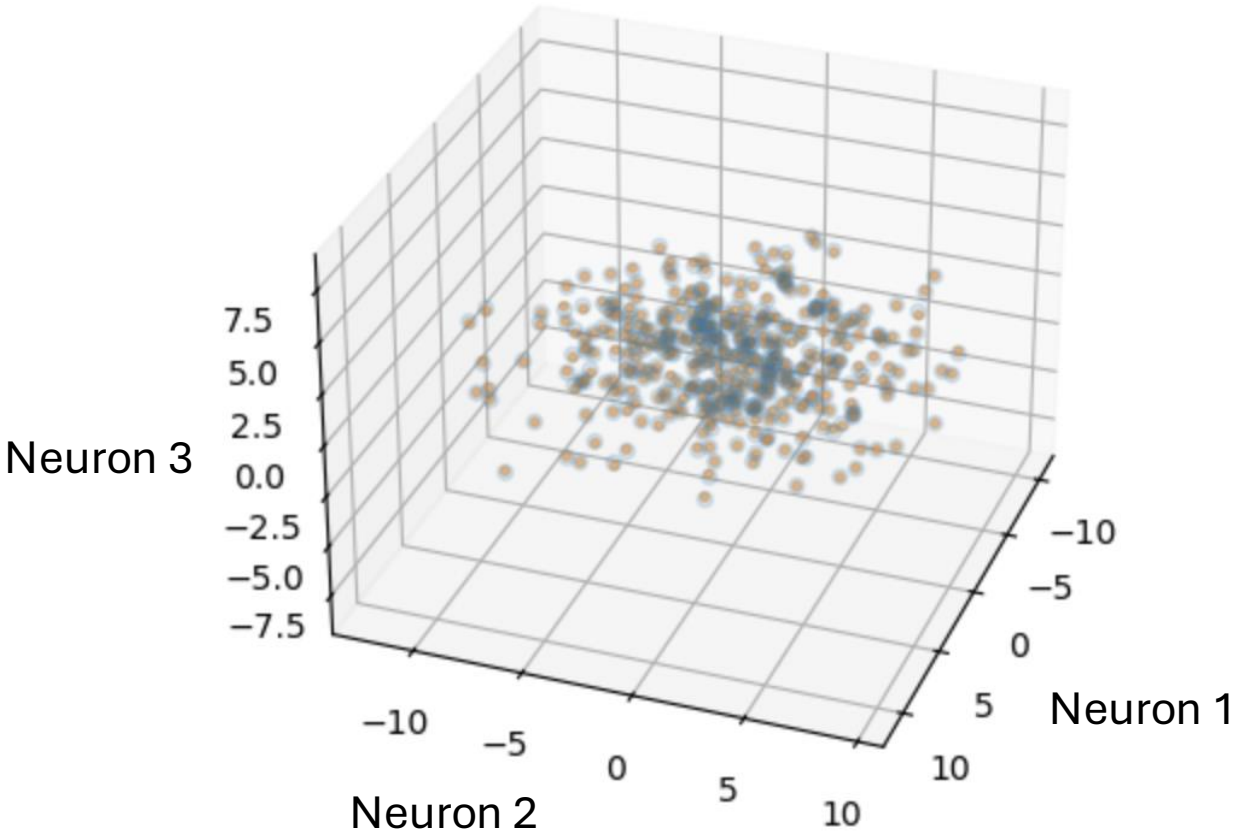

The data is very well reconstructed!

Cell 10 of notebook!

The first few principal axes can be used to reconstruct/denoise data!

Use only the first axis w_1:

$$\boxed{\hat{x}_c = \underbrace{(x_c^T w_1)}_{\text{Projection on } w_1} w_1}$$

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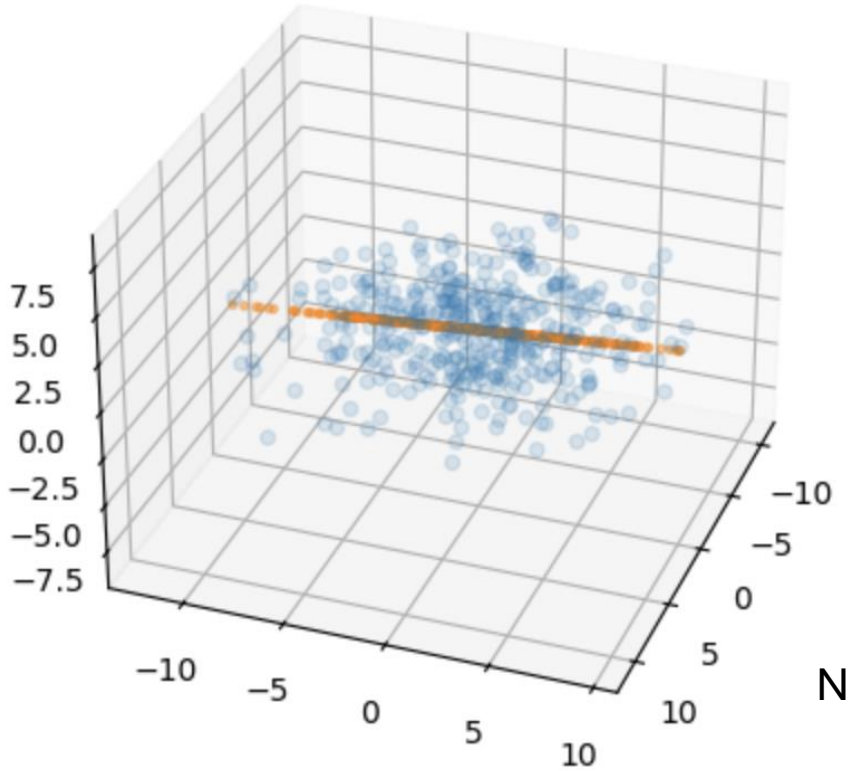

The data is only partially reconstructed.

Cell 11 of notebook!

Example PCA analysis on real neural data

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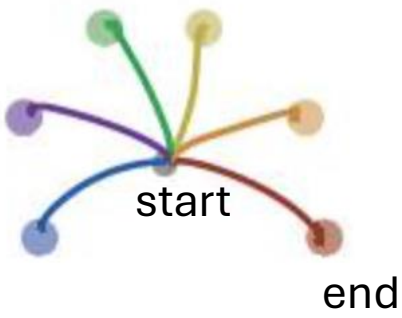
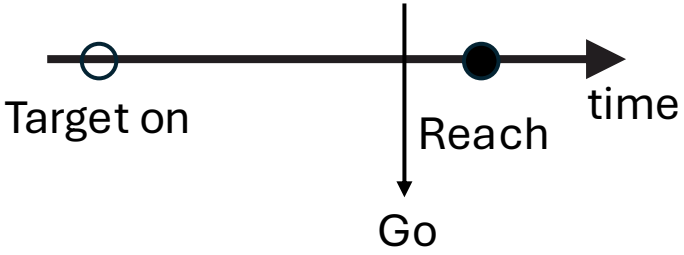
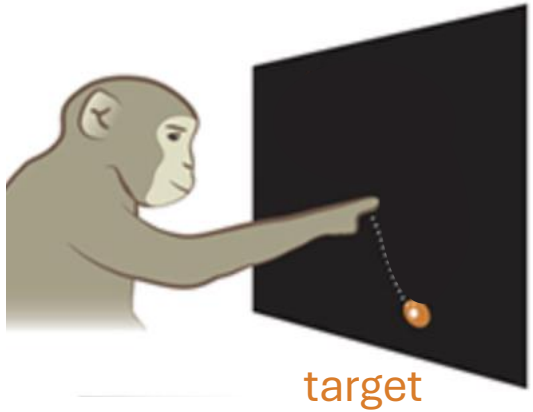


Arm trajectories





Zimnik and Churchland, 2021

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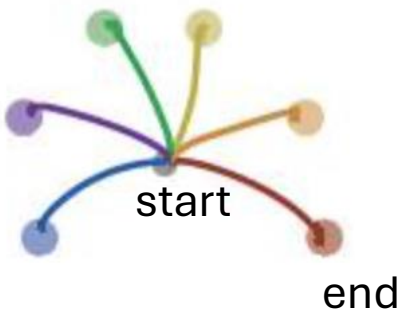

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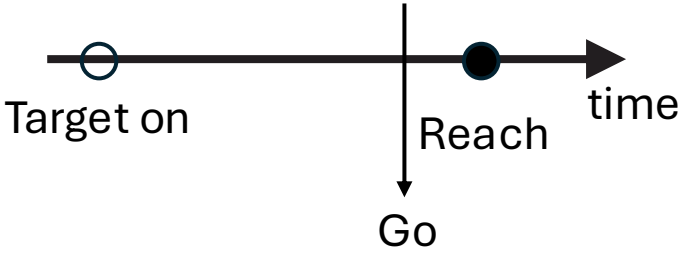

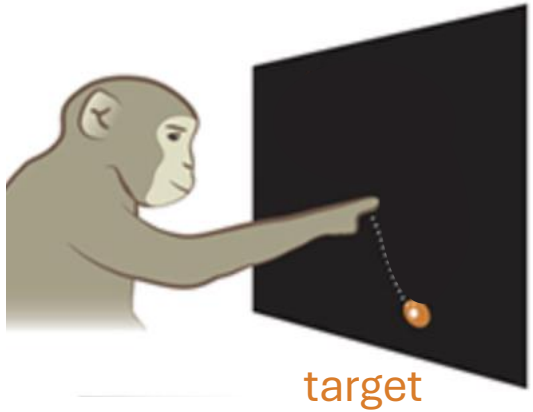

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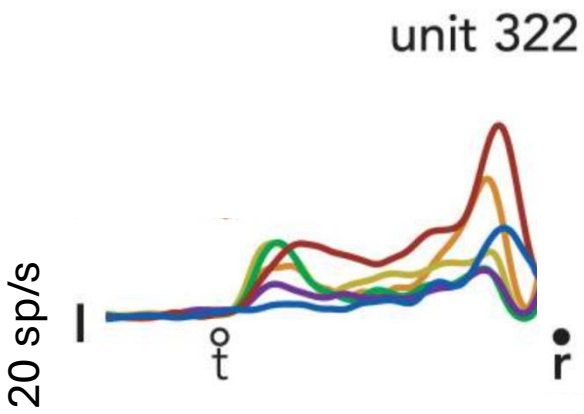
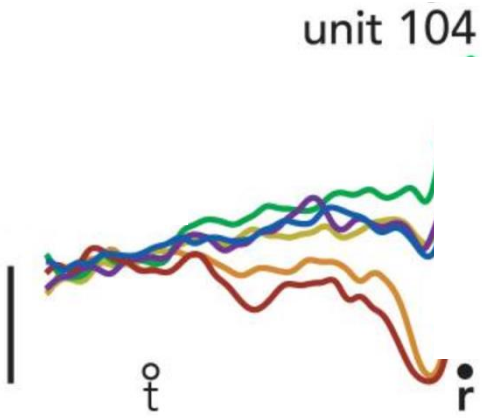
Single units:





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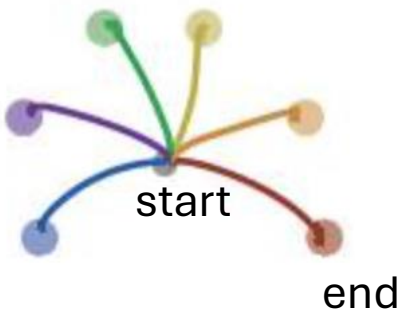

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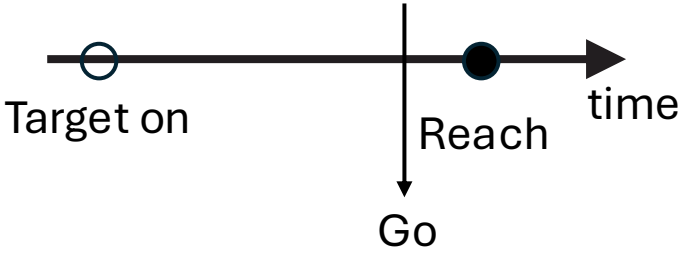

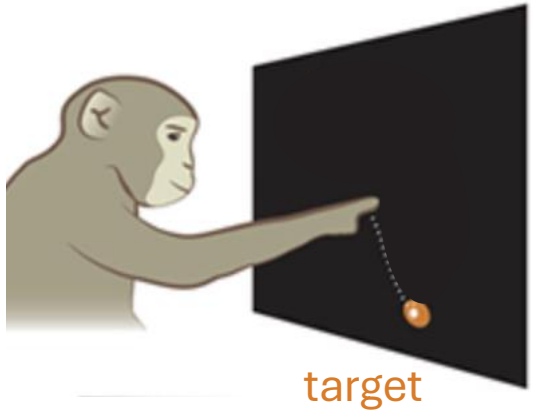

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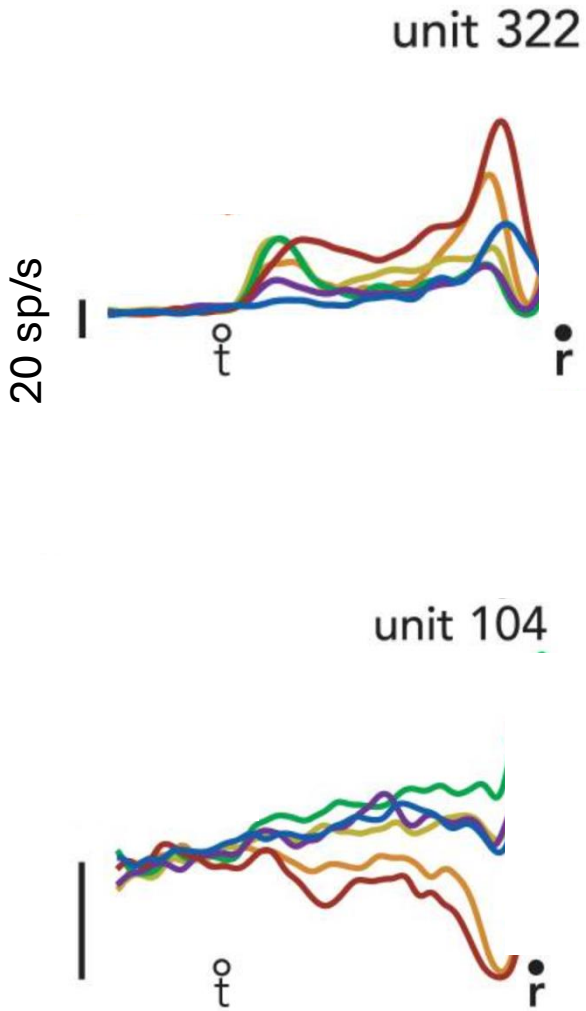
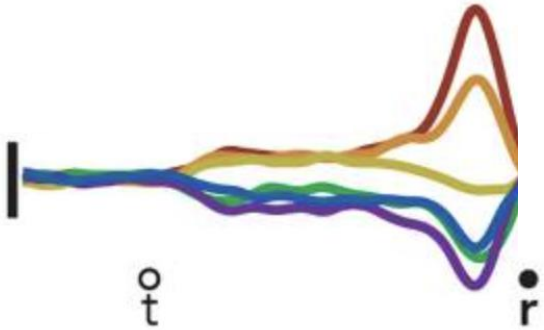


Projection on
first principal component



Method variations to extract a few ‘main’ components from data

- <u>Independent Component Analysis (ICA):</u>
  Finds statistically independent components (PCA finds uncorrelated ones).

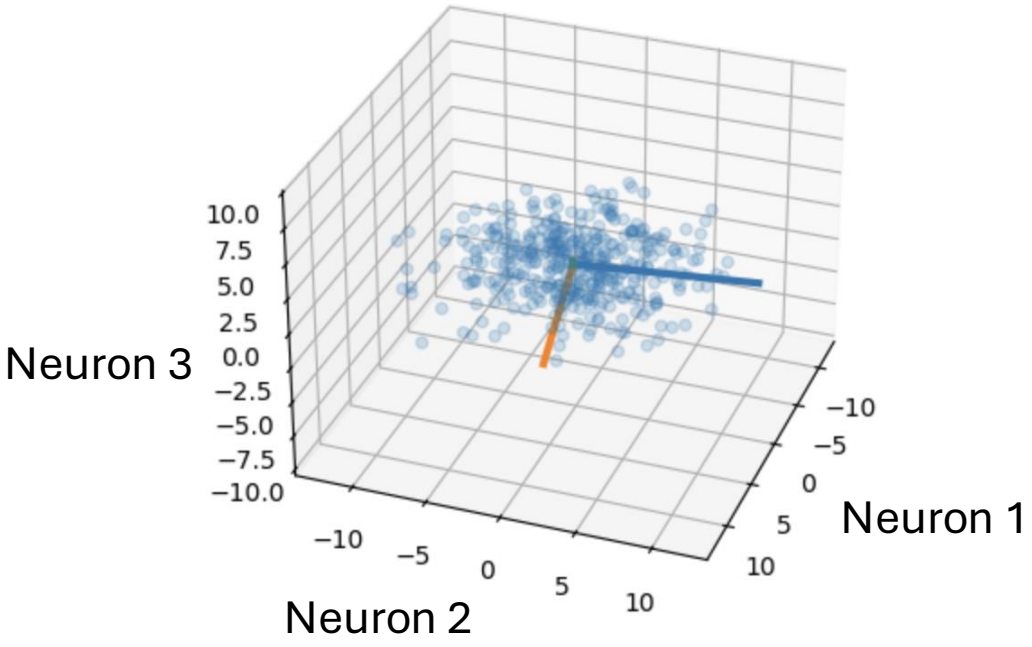
- <u>Demixed Principal Component Analysis (dPCA):</u>
  Optimizes both projection variance and separation of known labels.

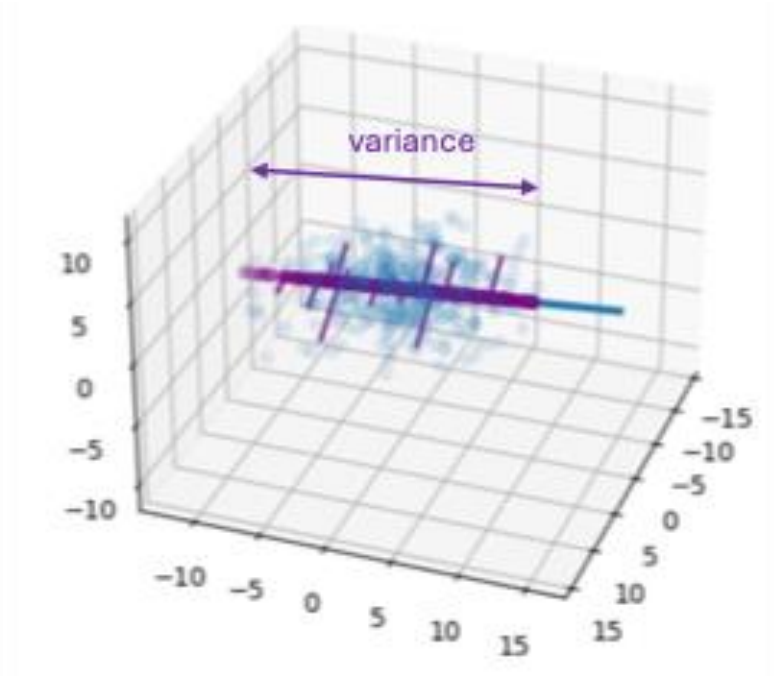
- <u>Factor analysis:</u>
  Models data as coming from a small set of underlying Gaussian distribution.

- <u>J Principal Component Analysis (jPCA):</u>
  Finds principal components and models data as a dynamical system creating rotations

• • •

Summary: Principal Component Analysis (PCA)





- Finds orthogonal directions – the principal axes – such that variance of the projected datapoints is maximized.
- Projection on the first few principal axes can be used to approximate, visualize and simplify data.
- If captured variance drops from principal axis n to n+1, it suggests axes >n may be negligible.